



Bahria University, Islamabad

Department of Software Engineering

Visual Programming LAB (Fall-2025)

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Project Purposal

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Project Proposal: EventSphere – A Social Event Networking Web Application

Project Overview

EventSphere is a modern, community-based web platform that allows users to create, join, and engage in social or professional events.

Every user in the system can act as both an **organizer** and an **attendee** — they can host their own events or participate in others' events seamlessly.

The application delivers a real-time, interactive experience where members can communicate through chat, share updates, and even host **live sessions** directly from the platform.

The project uses a **.NET Core Web API** for the backend and **React** with **Semantic UI** for the frontend, ensuring a structured, responsive, and visually consistent interface.

Core Features

1) User Registration & Authentication

- Secure user signup, login, and logout with encrypted credentials.
 - Each user maintains a personal profile that includes hosted and joined events.
 - Token-based session handling (JWT) for secure API communication.
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2) Event Creation & Management

- Any user can create events with details such as title, date, time, category, location, and description.
 - Event creators can upload a banner image or poster for their event.
 - Event hosts can view and manage participants within their own events.
 - Events can be edited or canceled by their creator.
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3) Event Discovery & Participation

- Browse all upcoming or active events in a clean, card-style interface.
- Search events by keywords, filter by category or date, and sort by newest or most active.
- Users can join or leave events instantly and see who else is participating.
- Joined events appear in a user's personal dashboard for quick access.

4) Real-Time Interaction

- Built-in **live chat** for every event, powered by **SignalR**.
- Messages appear instantly to all participants without reloading the page.
- Real-time notifications when a new message is posted or when users join an event.
- Presence indicators show active users within an event's chat session.

5) Live Sessions (Zoom Integration)

- Events can include an optional **live video session** powered by Zoom.
- **Organizers** (event creators) can start a live session directly from their event page.
- Attendees can join the Zoom meeting directly through a **Join Live** button.
- Two supported modes:
 - **Launch mode:** opens the Zoom meeting in a new tab using the join link.
 - **Embedded mode:** integrates the Zoom Meeting SDK within the website so users stay on the event page while watching the live session.
- The live session status (Live, Scheduled, or Ended) is displayed for clarity.

6) Event Insights Panel

- Each user has an overview section showing:
 - Events they have created or joined.
 - Participant counts and engagement levels.
 - Quick access to edit or start live sessions for their hosted events.
- Designed for personal event tracking and management — not an administrative control system.

7) Profile & Media Features

- Profile customization with image upload and basic details.
 - Image uploading and cropping features for event banners and profile pictures.
 - Image optimization for performance and clean presentation.
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Technical Details

<u>Category</u>	<u>Technology Used</u>
Frontend	React (JavaScript/JSX)
UI Library	Semantic UI React
Backend	ASP.NET Core Web API
Architecture	Clean Architecture with CQRS and Mediator pattern
Database	SQL Server (Entity Framework Core, Code-First)
Authentication	ASP.NET Identity with JWT
Real-Time	SignalR for chat and notifications
Video Integration	Zoom API / Zoom Meeting SDK
Mapping & Data Transfer	AutoMapper
Version Control	Git & GitHub
Deployment	Cloud-hosted, web-accessible platform

Expected Outcome

EventSphere will operate as a **fully functional social event network** where users can:

- Host and join events easily.
- Interact through real-time chat and notifications.
- Participate in or conduct live sessions from within the platform.
- Manage their personal event activities from a simple insights panel.

The final product will demonstrate a complete full-stack architecture integrating **.NET Core, React, Semantic UI, and SignalR** to create an engaging, community-driven web application.

Conclusion

EventSphere showcases the power of modern web development through real-time features, live event interaction, and seamless community engagement. By combining robust backend logic with a clean, Semantic UI-based frontend, the project provides an elegant and technically complete example of a dynamic event networking platform — one that allows every user to connect, create, and communicate effortlessly.